

MATA37H3 — Week 13

Taylor & Maclaurin Series: Analytic Functions

60 minutes (Pages 1–5, self-paced) • OpenStax *Calculus Volume 2*, §6.3 (Taylor and Maclaurin Series) — Theorem 6.6 (Uniqueness of Taylor Series), Theorem 6.7 (Taylor’s Theorem with Remainder), Theorem 6.8 (Convergence of Taylor Series), Table 6.1 (Maclaurin Series for Common Functions); §6.4 (Working with Taylor Series) — constructing new series by substitution/differentiation/integration. MAT137 Unit 14 slides (14.7–14.11: Lagrange’s Remainder Theorem and proving $\sin x$ is analytic; the general analyticity criterion; deriving $\arctan x$ as a power series and extracting a single high-order derivative value; “Taylor series gymnastics” — building new series from known ones) — MAT137 is the University of Toronto’s official exclusion-equivalent of MATA37H3.

DEFINITION Taylor Series, Maclaurin Series, Analytic Function

If f has derivatives of every order at $x = a$, the **Taylor series for f at a** is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n.$$

At $a = 0$ this is the **Maclaurin series**. A function f is **analytic at a** if there is an open interval containing a on which f equals its own Taylor series at a — i.e. the Taylor series doesn’t just converge, it converges to f itself. (A Taylor series can converge everywhere and still fail to equal f off a single point — this definition is exactly the guard against that.)

KEY FACT Theorem 6.7: Taylor’s Theorem with Remainder

Let f be $(n + 1)$ -times differentiable on an interval I containing a . For $x \in I$, define the remainder $R_n(x) = f(x) - p_n(x)$. Then there exists c strictly between a and x such that

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} (x - a)^{n+1}.$$

Corollary (Lagrange bound). If there is a single constant M with $|f^{(n+1)}(t)| \leq M$ for every t in I (not just at c — that’s the whole point, since c is unknown), then

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x - a|^{n+1} \quad \text{for all } x \in I.$$

KEY FACT Theorem 6.8: Convergence of Taylor Series $\iff R_n(x) \rightarrow 0$

$f(x)$ equals its Taylor series at a , for a given x , **if and only if** $\lim_{n \rightarrow \infty} R_n(x) = 0$. Strategy to prove a function is analytic on all of I : find a bound M (possibly depending on the interval, but *not* on n) for $|f^{(n+1)}|$ on I , plug into the Lagrange bound, and show the resulting bound $\rightarrow 0$ as $n \rightarrow \infty$ — usually because it contains a factorial in the denominator, which beats any fixed power or exponential in the numerator.

KEY FACT Maclaurin Series for Common Functions (Table 6.1)

$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ ($|x| < 1$), $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ (all x), $\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$ (all x),
 $\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$ (all x), $\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}$ ($-1 < x \leq 1$), $\arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$ ($-1 \leq x \leq 1$).
These six are the toolbox for building new series — you are expected to *recognize* a disguised version of one of these rather than differentiate from scratch. (The binomial series joins this toolbox in Week 14.)

KEY FACT Constructing New Series from Known Ones

Three legal moves, all justified by uniqueness (Theorem 6.6) and Theorem 6.4 (Week 11): **(1) substitute** a function of x into a known series in place of the variable; **(2) differentiate or integrate** a known series term-by-term; **(3) combine algebraically** — add, subtract, or multiply two known series by a common factor of x^k . Whichever moves you use, the interval of validity is inherited from the original series (intersected, if you combine two series with different intervals) — always state it.

Worked Examples

EXAMPLE 1

proving $\sin x$ is analytic on all of $\mathbb{R}$, using the Lagrange bound

Show $\lim_{n \rightarrow \infty} R_n(x) = 0$ for every real x , where $R_n(x)$ is the remainder for the Maclaurin series of $\sin x$.

Solution. Every derivative of $\sin x$ is $\pm \sin x$ or $\pm \cos x$, so $|f^{(n+1)}(t)| \leq 1$ for **every** $t \in \mathbb{R}$ and **every** n — a single bound $M = 1$ works on the whole real line, independent of n . By

the Lagrange bound, $|R_n(x)| \leq \frac{1}{(n+1)!} |x|^{n+1}$. Fix any x ; then $\frac{|x|^{n+1}}{(n+1)!} \rightarrow 0$ as $n \rightarrow$

∞ (factorial growth eventually dominates any fixed power). By Squeeze, $R_n(x) \rightarrow 0$ for every x . By Theorem 6.8, $\sin x$ equals its Maclaurin series for all $x \in \mathbb{R}$ — $\sin x$ **is analytic everywhere**. ■

EXAMPLE 2

the general analyticity criterion

State a hypothesis on f guaranteeing f is analytic on all of an interval I .

Solution. It suffices that there exist a **single constant** M (not depending on n) such that $|f^{(n+1)}(t)| \leq M$ for **every** n and **every** $t \in I$. Then exactly the argument of Example 1 applies verbatim: the Lagrange bound gives $|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1} \rightarrow 0$ for every $x \in I$, so $f =$ its Taylor series on all of I . *Misconception flag:* it is **not** enough for each individual derivative to merely be bounded (i.e. a possibly-different bound M_n for each n) — if M_n grows too fast with n , the ratio $M_n/(n+1)!$ need not $\rightarrow 0$. The bound must be **uniform in n** . ■

EXAMPLE 3

deriving $\arctan x$ as a power series, then extracting one derivative value

Given $G(x) = \arctan x$, derive its Maclaurin series and find $G^{(137)}(0)$.

Solution. $G'(x) = \frac{1}{1+x^2} = \frac{1}{1-(-x^2)} = \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n}$ for $|x| < 1$.

Integrate term-by-term: $G(x) = C + \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$; since $G(0) = \arctan 0 = 0$,

$C = 0$. So $\arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$. **To find $G^{(137)}(0)$** , don't differentiate 137 times

— match coefficients with the *definition* of the Maclaurin series: the coefficient of x^{137} in $\sum \frac{G^{(n)}(0)}{n!} x^n$ is $\frac{G^{(137)}(0)}{137!}$. Since $137 = 2n+1 \Rightarrow n = 68$ (and 137 is odd, matching the

series' odd powers, with sign $(-1)^{68} = 1$), the coefficient of x^{137} is $\frac{1}{137}$. Setting equal:

$$\frac{G^{(137)}(0)}{137!} = \frac{1}{137} \Rightarrow G^{(137)}(0) = \frac{137!}{137} = 136!. \blacksquare$$

EXAMPLE 4

exact evaluation of a numerical series, recognizing it as a known series value

Evaluate $A = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)3^n\sqrt{3}}$ exactly.

Solution. Recall $\arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$. We want the sum to look like this with x^{2n+1} producing a 3^{-n} factor: try $x = \frac{1}{\sqrt{3}}$, so $x^{2n+1} = \frac{1}{3^n\sqrt{3}}$ — matches exactly. So

$A = \arctan \frac{1}{\sqrt{3}} = \frac{\pi}{6}$ (a standard reference angle). *Misconception flag:* the temptation is to try to “compute” an infinite sum numerically — the entire point of this technique is to recognize the numerical series as a **known function's Maclaurin series evaluated at a specific point**, giving an exact closed form. ■

EXAMPLE 5

series gymnastics — building new series from known ones, several moves at once

Find the Maclaurin series for $f(x) = x^2 \cos x$ and state its interval of validity.

Solution. Start from $\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$ (all x). Multiply through by x^2 (a legal algebraic combination — pure substitution of the extra factor):

$$x^2 \cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+2}}{(2n)!},$$

valid for all x (multiplying by a polynomial doesn't change the radius of convergence). ■

EXAMPLE 6

series gymnastics — a rational function via geometric-series substitution and scaling

Find the Maclaurin series for $f(x) = \frac{x}{3+2x}$.

Solution. $\frac{x}{3+2x} = \frac{x}{3} \cdot \frac{1}{1+\frac{2x}{3}} = \frac{x}{3} \sum_{n=0}^{\infty} \left(-\frac{2x}{3}\right)^n = \sum_{n=0}^{\infty} \frac{(-1)^n 2^n}{3^{n+1}} x^{n+1}$, valid for $\left|\frac{2x}{3}\right| < 1 \iff |x| < \frac{3}{2}$. ■

Practice Problems

Read the hint and the caution before you start each problem — not after.

Proving analyticity

1. Show that e^x is analytic on every interval $[-R, R]$ (i.e. $R_n(x) \rightarrow 0$ for every fixed x), using the Lagrange bound with $M = e^R$ (justify why this M works for every $t \in [-R, R]$).

HINT Use the fact that e^t is increasing, so its maximum on $[-R, R]$ occurs at $t = R$, to justify the bound $M = e^R$.

CAUTION — before you solve The Lagrange bound requires a single constant M that works for every t in the interval — check that e^R really does dominate e^t for all $t \in [-R, R]$, not just at one point.

Your working:

2. Explain, in a sentence, why the argument in Problem 1 does **not** immediately give “analytic on all of $\mathbb{R}$ with a single bound M ” the way $\sin x$ ’s argument did — and why it doesn’t need to (what do you do instead to conclude e^x is analytic everywhere)?

HINT Notice that the bound $M = e^R$ depends on R itself — think about what changes if you let $R \rightarrow \infty$ versus fixing x first and choosing R afterward.

CAUTION — before you solve ‘Analytic on every bounded interval’ is not weaker than ‘analytic on all of $\mathbb{R}$ ’ — together they are the same statement, since every real number lies inside some $[-R, R]$.

Your working:

Deriving series and extracting a single derivative value

3. Given $f(x) = \frac{1}{1-x^3}$, write its Maclaurin series (substitution into the geometric series) and find $f^{(15)}(0)$.

HINT Substitute x^3 directly into the geometric series $\frac{1}{1-x} = \sum x^n$, then match the coefficient of x^{15} to $\frac{f^{(15)}(0)}{15!}$.

CAUTION — before you solve Most coefficients in this series are zero (only multiples of 3 appear) — don’t assume a derivative is ‘undefined’ just because its power isn’t a multiple of 3.

Your working:

4. Find $A = \sum_{n=0}^{\infty} \frac{1}{n!}$ exactly, by recognizing it as a known Maclaurin series evaluated at a specific point.

HINT Recognize the sum's pattern $\frac{1}{n!}$ as a known Maclaurin series evaluated at a specific x -value.

CAUTION — before you solve Don't try to compute this as an unfamiliar numerical series — recognizing which of the six core series (and at what x) it matches is the entire point of the technique.

Your working:

5. Evaluate $B = \sum_{n=0}^{\infty} \frac{(-1)^n \pi^{2n+1}}{4^{2n+1} (2n+1)!}$ exactly, by recognizing it as a known Maclaurin series value. (Hint: which of the six core series has odd powers over $(2n+1)!$ with alternating sign?)

HINT Look for the signature $(2n+1)!$ in the denominator with alternating sign — that identifies which of the six core series this is, then match the argument to $\pi/4$.

CAUTION — before you solve Don't confuse this with the $\arctan x$ series — $\arctan x$ has $(2n+1)$ in the denominator with no factorial, while this series has $(2n+1)!$.

Your working:

Series gymnastics

6. Find the Maclaurin series for $f(x) = e^{-x}$ and state its interval of validity.

HINT Substitute $-x$ directly into the known Maclaurin series for e^x .

CAUTION — before you solve e^{-x} is not the negative of e^x 's series — substitute term-by-term rather than trying to relate the two functions algebraically first.

Your working:

7. Find the Maclaurin series for $f(x) = \frac{e^x + e^{-x}}{2}$ (hyperbolic cosine) by adding two known series and dividing by 2. What happens to the odd-power terms, and why does that make sense given f is an even function?

HINT Add the e^x and e^{-x} series term-by-term and simplify $\frac{1+(-1)^n}{2}$ for even versus odd n .

CAUTION — before you solve Connect the cancellation of odd-power terms to the general even/odd function rule — an even function's Maclaurin series can only contain even powers, by symmetry, not by coincidence.

Your working:

8. Find the Maclaurin series for $f(x) = \ln \frac{1+x}{1-x}$ using $\ln(1+x) - \ln(1-x)$ and two applications of the known $\ln(1+x)$ series (careful with the substitution in the second one).

HINT Substitute $-x$ into the $\ln(1+x)$ series to get the $\ln(1-x)$ series, tracking $(-1)^{n+1}$ and $(-1)^n$ as separate factors before combining.

CAUTION — before you solve $(-1)^{n+1}(-x)^n$ simplifies to a constant sign, not an alternating one — track the two sign factors separately before multiplying them together.

Your working:
